

GRACE: A Tunable Metric Unifying Accuracy and Efficiency for Robotics Evaluation

Keisuke Sakaguchi*

Abstract—Robotics evaluation metrics generally report success rate (SR) and speed separately, ignoring the trade-off between them. To address this, we introduce **GRACE** (Generalized Robotics Accuracy-Completion Evaluation), a metric that unifies success rate and time efficiency through a tunable harmonic mean. The β parameter balances accuracy against speed, and the τ parameter controls deadline strictness. To validate GRACE, we rescore published results from Habitat ObjectNav, an RLBench generalization, and CALVIN ABC→D, comparing GRACE with weighted sum, SR/time, and success weighted by path length (SPL). The results show that GRACE can yield ranking reversals depending on the trade-off configuration between accuracy and efficiency while providing an interpretable parameterization. We release GRACE as an open-source toolkit to facilitate consistent evaluation of accuracy/efficiency across navigation, manipulation, long-horizon, and other robotics tasks.

I. INTRODUCTION

As physical AI expands across diverse domains, including warehouse automation, assistive care, disaster response, and robotic surgery, evaluation metrics vary accordingly, reflecting different priorities such as accuracy and efficiency. This highlights the importance of *generalizable systems* that can adapt to novel tasks without domain-specific engineering. Examples include RT-2 [1], which transfers vision–language knowledge to robotic control, PaLM-E [2], an embodied multimodal LLM, and the Open X-Embodiment project [3], which aggregates data across more than twenty robot types. Such systems shed light on the challenge of evaluation: how should we compare methods designed not for a single benchmark but for multiple tasks? How can we measure progress fairly across tasks with different goals, runtimes, and constraints? A fair comparison must consider success as well as efficiency, but they are typically reported separately, leaving the trade-off implicit.

To fill this gap, we introduce Generalized Robotics Accuracy-Completion Evaluation (GRACE), a tunable metric that makes the accuracy/speed trade-off explicit in robotics evaluation. GRACE combines accuracy and timing efficiency via a tunable harmonic mean based on the F_β score, which has been used in information retrieval research. Here, *runtime* denotes total wall-clock time from task initiation to completion, including planning and execution. Two parameters, β (accuracy/speed balance) and τ (deadline strictness), make domain assumptions explicit. Existing metrics are treated as

special cases of GRACE: success rate and pure efficiency are recovered at $\beta \rightarrow 0$ and $\beta \rightarrow \infty$.

This paper makes three main contributions:

- 1) We define GRACE as a tunable harmonic-mean score unifying accuracy and speed, and show how β and τ make domain values and time sensitivity explicit.
- 2) We conduct case studies on Habitat ObjectNav, an RLBench generalization sub-benchmark, and CALVIN, comparing GRACE against weighted sum, SR/ r , and SPL where applicable, and showing ranking reversals hidden by success-only reporting.
- 3) We release GRACE as an open-source toolkit at <https://github.com/tohoku-nlp/grace>, requiring only success and normalized time/cost inputs.

II. RELATED WORK

Benchmarks and platforms. Simulation benchmarks span navigation, manipulation, and long-horizon control. Habitat [4] provides photorealistic 3D navigation with success and path-length metrics, while AI2-THOR [5], RoboTHOR [6], and ProcTHOR [7] extend to manipulation and procedural variation, and Gibson [8] and iGibson 2.0 [9] add real-world scans and higher-fidelity physics. Manipulation and long-horizon benchmarks include RLBench [10], Meta-World [11], robosuite [12], and the DeepMind Control Suite [13], while BEHAVIOR-1K [14], ALFRED [15], ALFWORLD [16], and TEACH [17] target household and language-conditioned tasks; grasping datasets such as Dex-Net 2.0 [18] and tactile simulators such as TACTO [19] emphasize grasp robustness and contact quality. These platforms support reproducible evaluation, but report success and efficiency separately (e.g., success rate and SPL), with runtime treated as a secondary statistic.

Generalization and foundation models. Robustness studies and generalist policies also report accuracy alone. The Colosseum benchmark [20] perturbs environments to study generalization, and domain and dynamics randomization [21, 22] support sim-to-real transfer but report accuracy degradation, not efficiency; the speed–robustness interaction is rarely measured. Generalist policies such as RT-1/2 [1, 23] and Open X-Embodiment [3] similarly report success-rate aggregates with timing as secondary; none reports a combined accuracy/efficiency score.

Evaluation metrics. SPL (Success weighted by Path Length) [24], widely used in navigation, weights success by path efficiency:

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{\ell_i}{\max(p_i, \ell_i)}, \quad (1)$$

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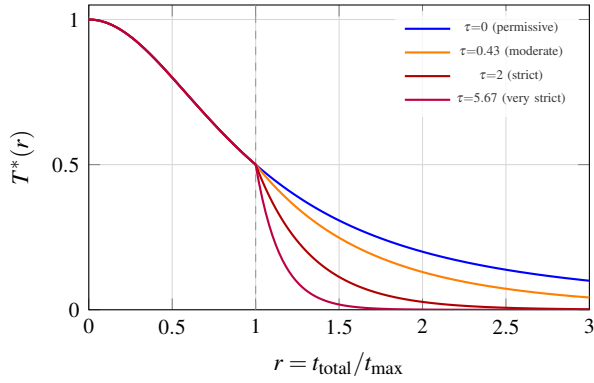


Fig. 1: **Normalized real-time ability $T^*(r)$ for different deadline strictness τ .** All curves decrease monotonically with r , so faster is always better. Beyond the deadline ($r > 1$), the exponential penalty controlled by τ determines how quickly T^* collapses. At $\tau=0$, only the $1/(1+r^2)$ base factor applies; at $\tau=5.67$, overruns are nearly fatal.

where $S_i \in \{0, 1\}$ is success, ℓ_i the shortest-path length, and p_i the agent’s path length for episode i . SCT (Success weighted by Completion Time) [25] extends SPL by replacing shortest path with fastest completion time under the agent’s dynamics model. Structurally, SCT is a per-episode product (success $\times$ optimal-time/actual-time), and it requires each episode’s instance-specific optimal completion time. GRACE’s normalizer is flexible: the same score accepts a fixed benchmark deadline when only aggregate numbers are available, and it can equally take a per-instance optimal completion time when available.

III. PROPOSED METHOD

We formalize Generalized Robotics Accuracy-Completion Evaluation (GRACE) and analyze its properties. The design adapts the F_β score from information retrieval [26] to robotics by incorporating a normalized runtime or cost signal.

A. Preference weights and internal mapping

To keep configuration intuitive, task designers (i.e., benchmark providers) specify two preference weights (parameters) $w_\beta, w_\tau \in [0, 1]$. These are mapped to the internal parameters $\beta > 0$ and $\tau \in [0, \infty)$ as

$$\beta = \sqrt{\frac{w_\beta}{1-w_\beta}}, \quad \tau = \frac{w_\tau}{1-w_\tau}. \quad (2)$$

Here β determines the balance between accuracy and speed ($\beta=1$ at $w_\beta=0.5$), while τ specifies deadline strictness ($\tau=0$ is permissive, larger τ imposes stricter penalties, and $\tau \rightarrow \infty$ approaches a hard cutoff). This parameterization makes GRACE flexible: varying (β, τ) spans preferences from accuracy-focused to speed-critical, with familiar metrics recovered as special cases—success rate only as $\beta \rightarrow 0$ and pure efficiency as $\beta \rightarrow \infty$.

B. Normalized runtime/cost and efficiency

In the timing setting, runtime t_{total} denotes the total wall-clock time from task initiation to completion (including planning, inference, and execution). It is normalized by the benchmark deadline t_{max} :

$$r = \frac{t_{\text{total}}}{t_{\text{max}}}, \quad r \geq 0. \quad (3)$$

The same equations can also be used with an explicitly declared normalized cost proxy; in that case, $r=1$ should be read as the chosen unit-cost anchor rather than a measured wall-clock deadline. We define the normalized efficiency term as

$$T^*(r) = \frac{1}{1+r^2} \exp(-\tau \max(0, r-1)). \quad (4)$$

We denote it by $T^*(r)$ to emphasize that it is a normalized and penalized efficiency term, rather than raw runtime or raw cost. The base factor $1/(1+r^2)$ provides $T^*(0)=1$ and $T^*(1)=0.5$ (a natural midpoint at the deadline). The quadratic form has zero derivative at $r=0$, so minor runtime differences among fast methods are not over-penalized, while the penalty steepens near the deadline; a linear alternative $1/(1+r)$ would penalize uniformly across all r . The exponential factor penalizes values beyond the unit anchor $r=1$, controlled by τ : as τ increases, the drop beyond $r=1$ becomes sharper, approaching a hard cutoff. Figure 1 illustrates $T^*(r)$ for four representative τ values.

C. The GRACE metric

GRACE is a weighted harmonic mean of success rate A and the efficiency term $T^*(r)$, adapting the F_β score. Unlike an arithmetic mean, which a single high factor can inflate, the harmonic mean requires *both* factors to be high; β sets their relative weight ($\beta=1$ weights them equally, $\beta > 1$ emphasizes efficiency). We define

$$\text{GRACE}(A, r; \beta, \tau) = \frac{(1+\beta^2)AT^*(r)}{\beta^2A + T^*(r)}, \quad (5)$$

where $A \in [0, 1]$ denotes the success rate.

Calibrating τ . Practitioners can calibrate τ from a policy statement: “20% overtime should halve credit” yields $\tau = \ln 2 / 0.2 \approx 3.47$; the paper’s strict regime ($\tau=5.67$) reduces the exponential overrun multiplier to 0.32 at 20% overtime.

D. Representative regimes

To show how domain preferences differ, Figure 2 shows GRACE isocontours in (r, A) space for four representative regimes: Household ($\beta=0.5, \tau=0$), Warehouse ($\beta=1, \tau=0.43$), Disaster ($\beta=2, \tau=5.67$), and Medical ($\beta=2, \tau \rightarrow \infty$). Table I provides an illustrative numerical example with four regimes that vary both β and τ . One model (M2) has the highest accuracy ($A=0.95$) but is slightly beyond the unit time anchor ($r=1.1$); another (M3) is the fastest ($r=0.3$) but least accurate ($A=0.60$). Under the accuracy-focused Household regime ($\beta=0.5$), M2 wins; under speed-critical Disaster or Medical regimes ($\beta=2$), M3 dominates and M2 collapses due to over-anchor penalties. This reversal shows how GRACE makes domain assumptions explicit.

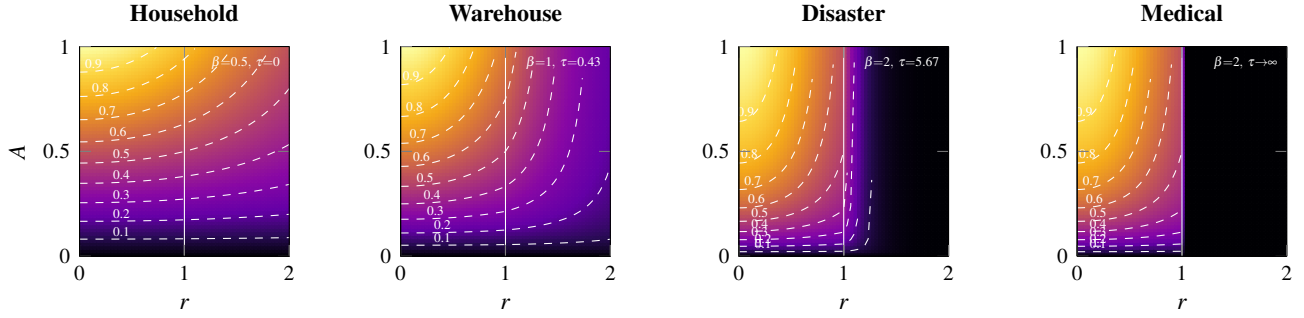


Fig. 2: **Representative time-sensitivity regimes in GRACE.** Each panel shows GRACE scores as a heatmap in (r, A) space; brighter regions indicate higher scores. White dashed lines are isocontours; the solid white line marks the deadline ($r=1$). Household ($\beta=0.5$): accuracy-dominated, score varies mainly with A . Warehouse ($\beta=1$): balanced trade-off. Disaster ($\beta=2, \tau=5.67$): steep penalty beyond $r=1$. Medical ($\beta=2, \tau \rightarrow \infty$): hard cutoff—any overrun yields zero. Brighter shading indicates a higher GRACE score (from 0 to 1).

TABLE I: **Illustrative domain-dependent GRACE scores.** Three hypothetical models with different (A, r) profiles are scored under four domains. HH (Household, $\beta=0.5, \tau=0$): accuracy-focused, permissive. WH (Warehouse, $\beta=1, \tau=0.43$): balanced, moderate. DS (Disaster, $\beta=2, \tau=5.67$): speed-critical, strict. MD (Medical, $\beta=2, \tau \rightarrow \infty$): speed-critical, hard cutoff. Colors indicate rank (1st, 2nd, 3rd). Rankings reverse across domains.

Model	A	r	HH	WH	DS	MD
M1	0.70	1.0	0.65	0.58	0.53	0.53
M2	0.95	1.1	0.78	0.60	0.30	0.00
M3	0.60	0.3	0.64	0.73	0.83	0.83

E. Properties

GRACE satisfies four key properties: (i) *Boundedness*: $\text{GRACE} \in [0, 1]$, with $\text{GRACE}=0$ when either $A=0$ or $T^*=0$, and $\text{GRACE}=1$ only when both $A=1$ and $r=0$; (ii) *Monotonicity*: GRACE is strictly increasing in both A and $T^*(r)$, so improving either dimension always improves the score; (iii) *Limiting cases*: $\beta \rightarrow 0$ recovers accuracy-only evaluation ($\text{GRACE} \rightarrow A$), $\beta \rightarrow \infty$ recovers efficiency-only ($\text{GRACE} \rightarrow T^*$), and $\tau \rightarrow \infty$ enforces a hard cutoff at the unit anchor; (iv) *Both-must-be-high*: as a harmonic mean, GRACE collapses when either factor is near zero, preventing compensation between success and efficiency.

IV. BENCHMARK RESCORING

Standardized, same-hardware wall-clock timing is not yet available for many robotics benchmarks. We therefore use case studies that make the status of the efficiency signal explicit. Habitat ObjectNav provides SR and SPL but not standardized timing, so we use a transparent path-efficiency proxy. The RLbench manipulation case uses the GravMAD generalization sub-benchmark, which reports both success and average task-completion time under one experimental setting. CALVIN ABC $\rightarrow$ D uses same-hardware per-action inference times reported together with completion performance. For each case, we compare GRACE against success rate only (SR), SR divided by normalized cost (SR/ r), a

weighted sum $\alpha A + (1-\alpha) \min(1, 1/r)$ at three α values, and, where available, SPL [24]. We evaluate three GRACE regimes: permissive ($\beta=1, \tau=0$), moderate ($\beta=1, \tau=0.43$), and strict ($\beta=2, \tau=5.67$).

A. Benchmarks and efficiency signals

Habitat ObjectNav (navigation). In ObjectNav, an agent is spawned in a previously unseen photorealistic indoor scene (HM3D) and must locate an instance of a target object category (e.g., *chair*, *bed*) from egocentric RGB-D and pose, succeeding if it stops near the object. Performance is summarized by success rate (SR) and Success weighted by Path Length (SPL). We use the HM3D ObjectNav validation comparison from SemNav [27], which reports SR and SPL for nine methods under a single setting: ZSON, ESC, VLFM, VoroNav, L3MVN, TriHelper, MFNP, SemNav-Greedy, and SemNav. Since per-method wall-clock runtime and step counts are not uniformly reported, we instantiate GRACE with a path-efficiency proxy derived directly from the published aggregates:

$$r_{\text{path}} = \frac{\text{SR}}{\text{SPL}} - 1. \quad (6)$$

Under the standard SPL definition, SPL/SR is the mean path efficiency among successful episodes. Thus r_{path} is the inverse conditional mean path efficiency minus one, a lower-bound proxy for fractional excess path length relative to the shortest path. Here $r_{\text{path}}=1$ corresponds to a conditional path efficiency of 0.5, i.e., paths roughly twice as long as shortest paths on average. It is not measured wall-clock runtime, and the Habitat rescoring should be read as a path-efficiency instantiation of GRACE rather than a timing experiment.

RLbench generalization sub-benchmark (manipulation). RLbench is a tabletop benchmark in which a simulated Franka arm executes language-specified, multi-step manipulation tasks (e.g., open a drawer, slide a block) from a handful of demonstrations, scored by task success rate. We use the GravMAD RLbench generalization setting rather than the canonical RLbench 18-task leaderboard, because the GravMAD paper reports both success and timing in one

TABLE II: **Benchmark rescoring with GRACE and baselines.** Published results are rescored with SR-only, SR/ r , weighted sum at three α values, published SPL for Habitat, and three GRACE regimes: permissive (P : $\beta=1, \tau=0$), moderate (M : $\beta=1, \tau=0.43$), and strict (S : $\beta=2, \tau=5.67$). Colors mark the 1st, 2nd, and 3rd best within each benchmark block and metric. Scores should be compared only within each block, because r denotes a different efficiency signal in each case.

			WSum									
Method	A	r	SR	$\frac{SR}{r}$	$\alpha=.3$	$\alpha=.5$	$\alpha=.7$	SPL	GRACE _P	GRACE _M	GRACE _S	
<i>Habitat ObjectNav (HM3D val; path proxy $r = SR/SPL - 1$)</i>												
ZSON	0.26	1.02	0.26	0.25	0.76	0.62	0.47	0.13	0.34	0.33	0.38	
ESC	0.39	0.76	0.39	0.52	0.82	0.70	0.57	0.22	0.48	0.48	0.57	
VLFM	0.53	0.73	0.53	0.72	0.86	0.76	0.67	0.30	0.58	0.58	0.62	
VoroNav	0.42	0.62	0.42	0.68	0.83	0.71	0.59	0.26	0.53	0.53	0.63	
L3MVN	0.50	1.18	0.50	0.43	0.74	0.68	0.61	0.23	0.46	0.44	0.17	
TriHelper	0.56	1.23	0.56	0.46	0.74	0.69	0.64	0.25	0.47	0.44	0.13	
MFNP	0.58	1.18	0.58	0.49	0.77	0.71	0.66	0.27	0.49	0.46	0.17	
SemNav-G	0.55	0.59	0.55	0.92	0.86	0.77	0.68	0.34	0.63	0.63	0.69	
SemNav	0.55	0.53	0.55	1.04	0.86	0.77	0.68	0.36	0.64	0.64	0.72	
<i>RLBench generalization (8 novel tasks; $r = \text{task time}/73.25\text{s}$)</i>												
VoxPoser	0.34	6.12	0.34	0.06	0.22	0.25	0.29	—	0.05	0.01	0.00	
Act3D	0.18	0.19	0.18	0.93	0.75	0.59	0.42	—	0.30	0.30	0.51	
3D-Diff.	0.29	0.68	0.29	0.44	0.79	0.65	0.51	—	0.41	0.41	0.54	
GravMAD	0.63	1.32	0.63	0.47	0.72	0.69	0.67	—	0.46	0.42	0.07	
<i>CALVIN ABC→D (A6000 inference time; $r = \text{latency}/14.4\text{ms}$)</i>												
Diff-P-CNN	0.27	0.81	0.27	0.33	0.78	0.64	0.49	—	0.37	0.37	0.48	
Diff-P-T	0.23	1.12	0.23	0.20	0.69	0.56	0.42	—	0.30	0.29	0.22	
RoboFlam.	0.49	4.51	0.49	0.11	0.30	0.36	0.41	—	0.09	0.02	0.00	
SuSIE	0.54	13.82	0.54	0.04	0.21	0.31	0.40	—	0.01	0.00	0.00	
GR-1	0.61	0.88	0.61	0.70	0.88	0.81	0.73	—	0.59	0.59	0.57	
MoDE	0.80	0.85	0.80	0.95	0.94	0.90	0.86	—	0.67	0.67	0.62	

source [28]. The setting trains on 12 base RLBench tasks and evaluates generalization on 8 novel RLBench tasks. We use the reported average success rates for VoxPoser, Act3D, 3D Diffuser Actor, and GravMAD (VLM), together with the reported average task-completion times from the same appendix, measured under the 8-novel-task setting on a single NVIDIA 4090 GPU. Because GRACE defines runtime at the task level, we use average task-completion time rather than keypose-only prediction time; note that task-completion time bundles motion planning and physical execution, not only policy inference, so it is a task-level cost rather than a pure compute measure. We normalize by the pool median task-completion time (73.25 s), so $r=1$ corresponds to the median method in this source.¹

CALVIN ABC→D (long-horizon). CALVIN is a long-horizon, language-conditioned tabletop benchmark in which an agent completes chains of up to five sequential sub-tasks given in natural language. The ABC→D split trains on environments A, B, and C and tests on the unseen environment D, measuring zero-shot generalization by the average number of consecutive subtasks completed (out of five). We use the CALVIN ABC→D split from MoDE [29]. Success is completion depth, $A = \text{Avg. Len.}/5$, for Diffusion Policy-CNN, Diffusion Policy-Transformer, RoboFlamingo, SuSIE, GR-1, and MoDE. The same source reports per-action inference time for all six methods on a single NVIDIA A6000 GPU over 100 runs. We normalize inference time by the pool median (14.4 ms), so $r=1$ marks median per-action

latency. This is our most direct same-hardware efficiency measurement, though it captures policy inference latency rather than full physical rollout time.

B. Results

Table II shows the full rescoring.

Real-benchmark ranking reversals. Habitat shows a path-efficiency reversal. MFNP and TriHelper rank first and second by SR, but their SPL-derived excess-path proxies exceed the unit path-cost anchor ($r=1.18$ and 1.23). SemNav, ranked third by SR but best by SPL-derived path efficiency ($r=0.53$), ranks first under all GRACE regimes. Under the strict regime, the more efficient SemNav-Greedy, VoroNav, and VLFM also outrank the higher-SR but less efficient MFNP and TriHelper.

RLBench shows a manipulation reversal under speed-critical settings. GravMAD has the highest success ($A=0.63$) but also longer average task-completion time ($r=1.32$). It ranks first under permissive and moderate regimes, where accuracy remains important, but drops under the strict regime; 3D Diffuser Actor becomes first and Act3D second because both complete tasks faster. GRACE changes the ranking only when the regime makes speed matter enough.

CALVIN shows a same-hardware inference-latency reversal below the top method. MoDE remains first because it is both the most successful and among the fastest. However, SuSIE and RoboFlamingo, which rank third and fourth by completion depth, drop to the bottom under GRACE because their inference latencies are much larger than the other policies. Diffusion Policy-CNN, though low by success, rises to third under GRACE because it is fast.

¹This is a within-paper timing case study, but it should not be interpreted as a rescoring of the standard RLBench 18-task benchmark.

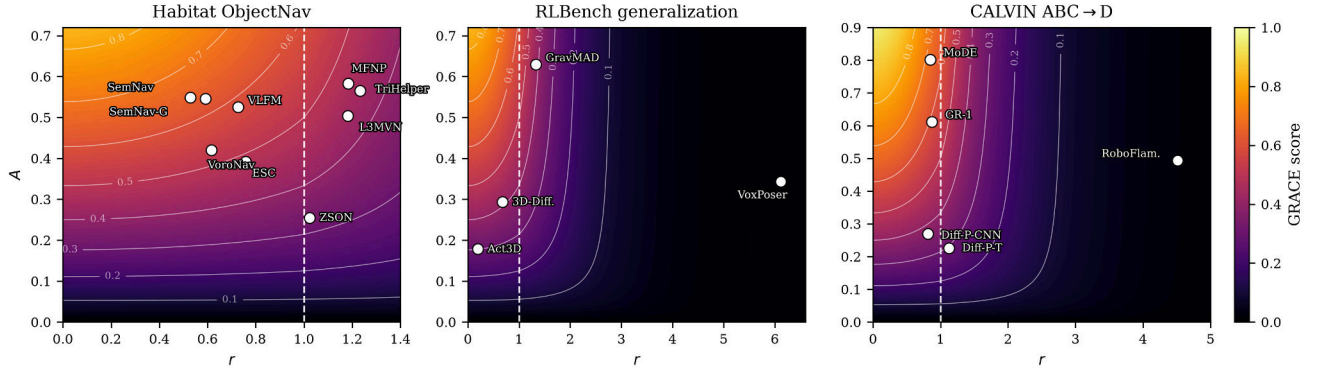


Fig. 3: **Benchmark case studies in (r, A) space** under the moderate GRACE regime ($\beta=1, \tau=0.43$). Brighter regions indicate higher scores and the dashed vertical line marks the unit anchor ($r=1$). Habitat uses an SPL-derived path-efficiency proxy; RLbench uses average task-completion time from the GravMAD generalization setting; CALVIN uses same-hardware per-action inference time. For readability, the CALVIN panel omits SuSIE ($r \approx 13.8$), whose large per-action latency would otherwise compress the axis and darken the whole panel; SuSIE is retained in all quantitative results (Table II).

Weighted-sum compensation. The weighted sum reorders methods as α changes and can allow one strong factor to mask a weak one. In RLbench, GravMAD retains a high $WSum_{0.7}$ score because of its success rate, even though its task completion time is above the normalized time anchor. In Habitat, MFNP and TriHelper remain near the top under $WSum_{0.7}$ despite their inefficient paths. The harmonic mean instead penalizes cases where either success or efficiency is low, and its parameters have direct semantics: β controls success/efficiency emphasis and τ controls above-anchor strictness.

Within- vs. above-anchor behavior. When $r < 1$ relative to the chosen runtime or cost anchor, the exponential penalty is inactive and GRACE still differentiates methods through the base factor $1/(1+r^2)$. When $r > 1$, τ determines how sharply above-anchor methods are penalized. This is why Habitat methods with long excess paths and RLbench/CALVIN methods with long completion or inference times fall sharply in the strict regime. SPL also rewards navigation efficiency, but it does not expose a tunable strictness parameter and is unavailable for manipulation and long-horizon benchmarks.

Rank correlations. Kendall’s τ_K between SR and moderate GRACE is +0.33 for Habitat, +0.33 for RLbench, and +0.33 for CALVIN. These low positive values reflect that the efficiency signal substantially reorders success-only rankings. For Habitat, permissive GRACE correlates strongly with published SPL ($\tau_K=0.83$), but the strict regime correlates less ($\tau_K=0.44$) because the strict above-anchor penalty reweights SPL-derived efficiency nonlinearly. These correlations are indicative rather than conclusive because each case study has only a small number of methods.

Parameter sensitivity. We sweep $\beta \in \{0.5, 1, 2, 3\}$ and $\tau \in \{0, 0.43, 2, 5.67\}$. Habitat and CALVIN have stable rank-1 methods across all 16 cells: SemNav leads Habitat because it combines high SR with the best path-efficiency proxy, and MoDE leads CALVIN because it is both the most accurate and among the fastest. RLbench is deliberately

TABLE III: **RLbench rank-1 method under different (β, τ) settings.** The ranking is intentionally regime-dependent: the most successful method (GravMAD) wins in accuracy-favoring settings, while progressively faster methods (3D Diffuser Actor, then Act3D) win as speed emphasis and strictness dominate. Habitat and CALVIN are omitted because their rank-1 method (SemNav and MoDE, respectively) is stable across all 16 cells.

β	τ			
	0	0.43	2.0	5.67
0.5	GravMAD	GravMAD	GravMAD	3D-Diff.
1.0	GravMAD	GravMAD	3D-Diff.	3D-Diff.
2.0	3D-Diff.	3D-Diff.	3D-Diff.	3D-Diff.
3.0	Act3D	Act3D	Act3D	Act3D

regime-dependent (Table III): GravMAD leads in accuracy-favoring settings, 3D Diffuser Actor leads in balanced-to-strict settings, and Act3D leads only when the regime is extremely speed-dominated. This is the intended behavior: the ranking changes exactly when the declared priorities change.

Geometric view. Figure 3 visualizes the methods in (r, A) space overlaid with moderate-regime GRACE isocontours. Habitat methods separate along the SPL-derived path-excess axis. RLbench methods show the accuracy-efficiency trade-off: GravMAD is more successful but slower, whereas Act3D is less successful but much faster. CALVIN shows that very slow policies such as RoboFlamingo and SuSIE are separated from fast policies even when their completion-depth scores are competitive.

V. DISCUSSION

The rescoring in Section IV shows that GRACE reveals trade-offs hidden by success-only reporting while making the status of each efficiency signal explicit. Habitat demonstrates a navigation case where SR and SPL can be converted into a transparent path-efficiency proxy. RLbench provides a ma-

TABLE IV: **Structural properties of evaluation metrics.**
✓ = satisfied, ~ = partially, ✗ = not satisfied.

Property	SR	WSum	SPL	GRACE
Uses timing/cost information	✗	✓	✓	✓
Differentiates within-deadline	✗	✗	✓	✓
Non-compensatory (both high)	✗	✗	~	✓
Deadline expressiveness	✗	✗	✗	✓

nipulation case where the ranking changes with the declared application regime: the high-success method wins in permissive settings, while faster methods win in speed-critical settings. CALVIN provides a same-hardware inference-latency case where slow policies drop despite strong completion-depth scores.

Why not a weighted sum? A weighted sum $\alpha A + (1-\alpha)\min(1, 1/r)$ has two structural limitations that GRACE addresses. *First*, the linear form allows one high factor to mask a low one. In RL Bench, GravMAD has the highest success but is above the normalized time anchor; a high- α weighted sum still keeps it near the top, whereas strict GRACE demotes it because the harmonic mean requires both success and efficiency to be high. *Second*, the weighted sum has no explicit deadline model. We use $\min(1, 1/r)$ because it matches common practice, but this term saturates for all within-deadline methods and cannot express different overrun policies. GRACE differentiates within the deadline through $1/(1+r^2)$ and penalizes overruns through τ . Table IV places these differences alongside SR and SPL across four structural properties. SPL also uses timing and differentiates methods within the deadline, but only partially resists compensation and offers no tunable deadline; the weighted sum uses timing yet saturates within the deadline and compensates freely; success rate ignores timing altogether. Only GRACE satisfies all four.

Mitigating parameter subjectivity. Because β and τ are tunable, a natural concern is that authors might select values that flatter their own method. Two practices reduce this freedom. First, report GRACE at fixed, domain-chosen regimes such as those in Section III and Table I, rather than tuning (β, τ) per paper. Second, always co-report the raw inputs A and r alongside the score. Readers can then recompute GRACE under any regime and immediately see whether a low score stems from poor success, poor efficiency, or both. **Generality of the success input.** GRACE defines success as a general $A \in [0, 1]$ that can encode any task-level quality criterion: grasp success, collision-free completion, navigation SR, or chain completion rate. In CALVIN, for example, we use $A = \text{Avg. Len.}/5$. GRACE is agnostic to how A is composed; it combines the benchmark-defined quality signal with a normalized time or cost signal.

Relation to SPL. SPL [24] weights success by path efficiency. For our Habitat rescoring, the efficiency proxy is itself derived from SPL (Eq. 6), so GRACE on Habitat is by construction a function of SR and SPL and is best read as a navigation-specific path-efficiency instantiation rather than an independent runtime measurement. Under the permissive

regime, GRACE tracks SPL closely; under the strict regime, it can diverge because τ adds an explicit above-anchor penalty that SPL does not model. Unlike SPL, the same GRACE form also applies to manipulation and long-horizon benchmarks when timing is available.

Limitations. Safety violations, human interventions, and domain-specific quality factors are not captured by the core metric; these remain optional extensions or can be folded into A . Our rescoring uses published aggregate numbers rather than per-episode data. Because GRACE is nonlinear, computing it from aggregate A and mean r is not equivalent to averaging per-episode scores. Per-episode logs with confidence intervals would provide stronger statistical claims and should be a priority for future benchmark releases.

VI. CONCLUSION

We introduced GRACE, a tunable harmonic-mean metric unifying success rate and timing efficiency for evaluating diverse robotics tasks. Two simple parameters, β (accuracy/speed balance) and τ (deadline strictness), make evaluation flexible across task configurations (e.g., accuracy-focused household robots vs. speed-critical disaster response). Case studies on Habitat ObjectNav, an RL Bench generalization sub-benchmark, and CALVIN ABC→D show that GRACE reveals trade-offs hidden by success-only reporting. To keep evaluation objective across the robotics community, we recommend reporting GRACE at fixed, domain-chosen regimes and always co-reporting the raw success A and normalized cost r . GRACE is released as an open-source toolkit to facilitate consistent evaluation across robotics tasks.

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